

Power Measurement and Control Using Programmable Logic

This article discusses measurement and control of power consumption in a complex multiple-rail supply system and proposes the use of programmable logic devices for the same

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In a complex system design, power consumption of hardware components is a very important performance metric because of the resulting thermal and reliability concerns. Reducing power consumption is a challenge to system designers. Portable systems such as laptops and personal digital assistants (PDAs) draw power from batteries, so reducing power consumption extends their operating time. For desktop computers or servers, high power consumption raises the temperature and deteriorates the performance and reliability.

Measuring and controlling power consumption in a multirail supply system is always tricky for the board designer. The major tasks of measuring, controlling and indicating power consumption should be accurate, easy to accomplish, flexible and scalable depending upon the application.

Power measurement and calculation

For an effective power management and control, the first step should be to accurately measure power consumption of the system.

Power = $V_{\text{supply}} \times I_{\text{supply}}$
where V_{supply} is the input voltage and I_{supply} the current drawn by the system.

Voltage (V_{supply}) can be easily detected by an operational amplifier or converted into digital by an analogue-to-digital converter (ADC). For measuring the current drawn by the supply (I_{supply}), one technique is to have a fixed resistor in series with the supply path and measure the voltage across this resistor—which is basically Ohm's law.

$$V_{\text{rs}} = I_{\text{supply}} \times R_{\text{sense}}$$

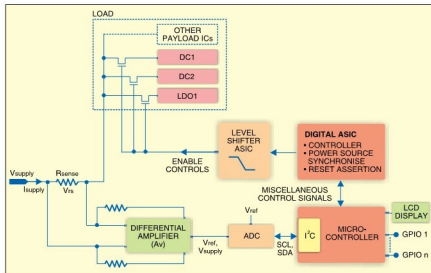


Fig. 1: A method to determine power consumption on the board

where V_{rs} is the voltage across the resistor and R_{sense} the value of the series resistor that can be easily detected.

The value chosen for R_{sense} is very small—typically less than one ohm—in order to reduce the voltage drop and power dissipation across it. Usually, V_{rs} would be a small voltage that can be amplified further. A differential or an instrumentation amplifier can amplify this voltage sufficiently to measure.

Now the power is given by:

$$\text{Power} = V_{\text{supply}} \times \frac{A_v(V_{\text{rs}})}{R_{\text{sense}}}$$

where A_v is the gain of the amplifier.

A general method of measuring power consumption is shown in Fig. 1.

This method uses a differential amplifier to provide gain on V_{rs} . It uses a two-channel discrete ADC to convert the analogue supply voltage (V_{supply}) and the amplified voltage across series resistor R_{sense} ($A_v(V_{\text{rs}})$) into digital quantities. The ADC resolution or the number of bits for each voltage determines the accuracy

of these conversions. A microcontroller then fetches this data and performs the required actions to calculate, control and indicate power. Additional ICs may be required to enable/disable other payload ICs on the board.

The microcontroller uses the two digital inputs and calculates power by implementing a digital multiplier and divider—for basically solving the equations—through its firmware. The power control mechanism to reduce the power consumption is implemented in the microcontroller. All other routine jobs like enabling controls and reset generation are done with additional digital or analogue ASIC and some glue logic within a discrete component.

There are a number of limitations in the above approach for measuring power:

1. In a multiple-rail system, scaling up or scaling down the design, depending on the number of rails, might require a board re-spin.
2. Flexibility of user-defined actions